MESA COHORT C4

AI and its Applications — Section 1 Final Examination

PROBLEM STATEMENT BOOKLET

Instructions to Students

1. Read your assigned problem statement carefully.
2. Your solution must use live external data wherever possible. The system must work with some data at the time of submission.
3. All AI tasks must use Gemini Flash. Do not use Claude API.
4. Deploy your solution on Vercel. Submit a working public URL along with your GitHub repository link and the Components Map.

High Difficulty

Podcast Ecosystem Radar

Creator Economy • Audio Content • SaaS-able

Aditya has been hosting a podcast for independent startup founders for two and a half years. He publishes two episodes a week and has built a committed audience of around nine thousand monthly listeners. The show's growth depends on two things: booking guests who are interesting and slightly ahead of the curve - founders working on something significant before they're inundated with podcast requests - and attracting brand sponsorships from companies that are actively spending in the Indian startup content space. Both require intelligence that he is currently getting too slowly and too randomly.

To find guests, Aditya relies on Twitter recommendations, listener suggestions, and one-off introductions from his network. He misses a lot. Founders who are doing media rounds across multiple podcasts right now - which signals they are available, media-willing, and in the middle of a public moment - often don't reach him until the moment is already fading. To find sponsors, he cold-pitches companies with no signal about who is actually spending on podcast advertising in

his space. He pitches companies that have never touched podcast budgets and misses the brands already buying shows just like his.

Aditya wants an intelligence system that monitors the ten podcasts in his content ecosystem (you can pick any 10) and surfaces what is happening across them each week. He wants to know which guests are appearing on multiple shows in the same month - because those are the people actively doing media rounds and easiest to book right now. He wants to know which topics are being covered across multiple shows simultaneously - because those are the themes to ride or counter-programme against. He wants to know which brands and products are being mentioned or promoted across shows in his space - because those are companies actively spending on podcast advertising. All three tell him where to focus: his booking outreach, his content calendar, and his sponsor pitch list. The set of shows he monitors should be configurable without touching the code, and the brief should arrive every Monday morning before he records.

> Build and deploy an AI-powered weekly podcast ecosystem intelligence system for Aditya, delivered every Monday morning before he starts his production week.